



Estimate Deflection

Bend the ski under a load case: an EI curve, up to three supports and two applied loads, each spread over a width -- out comes the deflected shape.

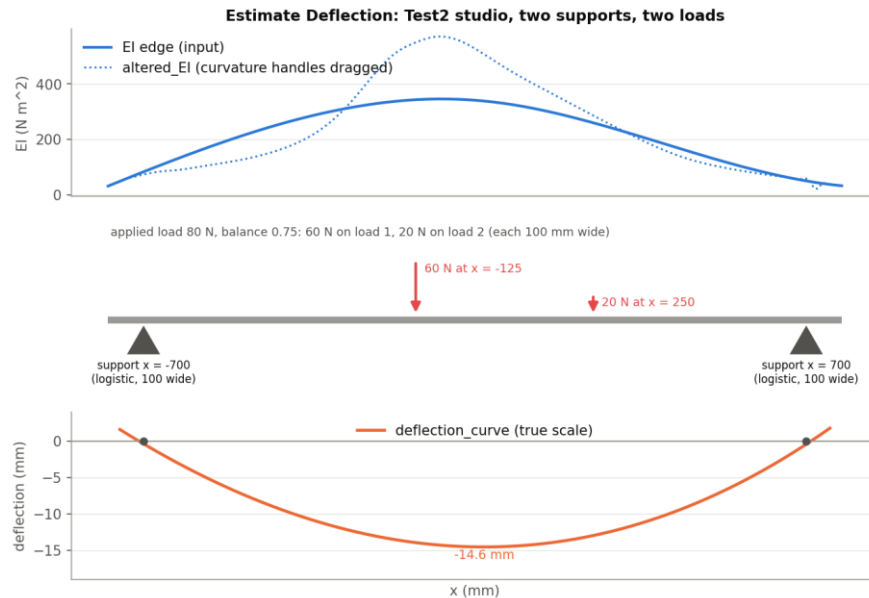


What it does

Flex numbers are one load case. Estimate Deflection solves any: a bench test, a skier's weight fore and aft, a binding plate. Curvature = M / EI integrated twice gives the shape, zero at supports 1 and 2.

Use it when

- Predict a bench or on-snow deflection
- See the effect of a binding plate or mounting moment
- Find the EI that would give a shape (drag the curvature)



Test2: supports at ± 700 mm, 60 N and 20 N loads $\rightarrow 14.6$ mm.



The dialog (1/2)

Estimate Deflection 1

✓

✗

1

El Edges
Edge of Measured_EI

(Q)

2

Number of evaluation points

100

3

☒ Show shear/moment diag...

4

☐ Alter deformed beam curv...
☐ Print load summary
☐ Print full diagnostics

#	Parameter	What it means
1	El Edges	The EI curve: an edge in the XZ plane, world Z in mm = EI in N m^2 (EI and Cross Section's <name>_EI, a measured or a target curve).
2	Number of evaluation points	20-500 (100); also EI samples per edge.
3	Show shear/moment diagrams	Draw V and M.
4	Alter deformed beam curvature	Drag curvature handles; outputs the matching EI (altered_EI).



The dialog (2/2)

5 ☒ Add plate/mounting con...

Plate stiffness (Q)

Reaction moment 0

Moment x location 10 mm

6 ☒ Applied load data

Applied load 80

Applied load balance 0.75

☒ Applied load 1

X_VAL

Applied load 1 load shape

QUINTIC

Applied load 1 width 25...

Applied load 1 x loc... -1...

☒ Second applied load?

☒ Applied load 2

Applied load 2 Location T...

X_VAL

Applied load 2 shape

CONSTANT

Applied load 2 width 25 ...

Applied load 2 x loc... 2...

7 ☒ Support load data

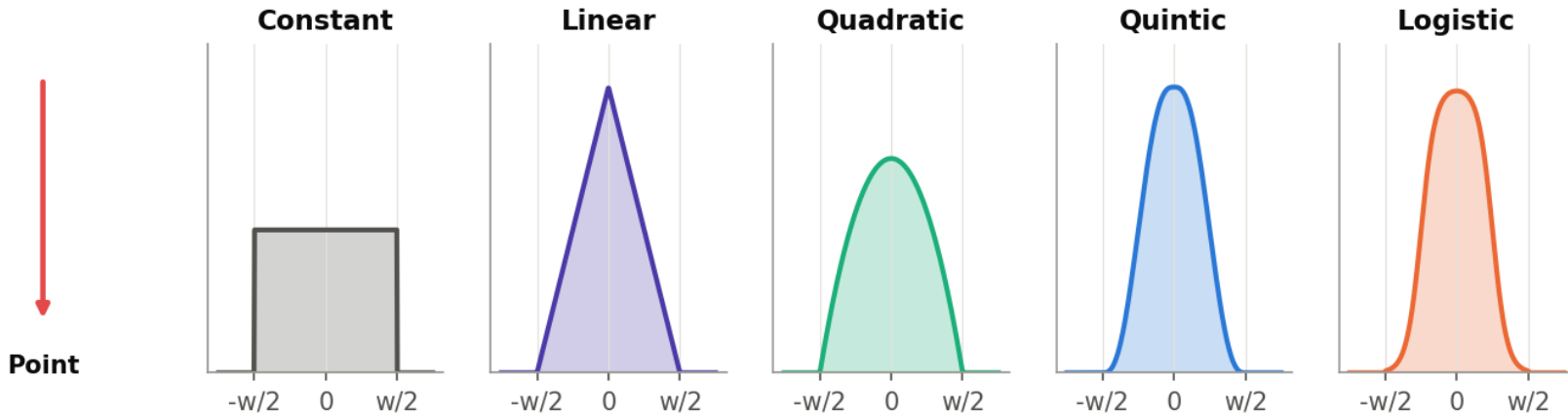
Final

#	Parameter	What it means
5	Add plate/mounting conditions?	Plate stiffness edge, reaction moment (N m, positive) and its x.
6	Applied load data	Load 3-600 N, balance between two loads; each: location, shape, width.
7	Support load data	Supports 1, 2 (optional 3 with its share); each: location, shape, width.
8	Output Options	Span (full EI / supports); Fit or Approximate.



Load shapes

Load shapes as designed: each spreads the force over the width w (area = force)



Spread over a width

Each shape integrates to the load's force.

Supports too

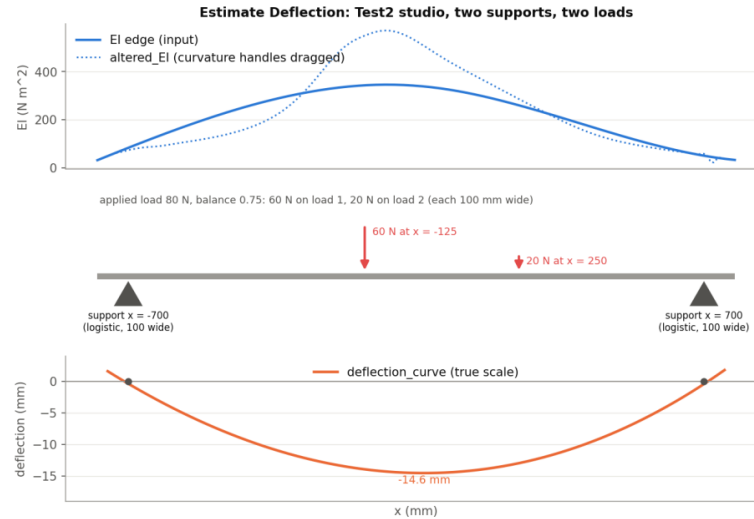
Reactions can be spread (Logistic suits a support pad).

Known bug

Every distributed shape is computed as Logistic today (enum compared with strings). Point is correct.



Example: Test2



EI eiTarget; supports x = -700 / 700; 80 N split 0.75: 60 N at -125, 20 N at 250 (100 mm wide)

The deflection_curve sags 14.6 mm near the middle. With 'Alter deformed beam curvature' on, the dragged shape gives altered_EI (dotted).



Outputs and messages

Outputs

Key	What it holds
deflection_curve	The deflected shape, Z = deflection, true scale.
altered_EI	Optional: the EI curve that gives the dragged curvature (EI format).

Messages

	Message / what to do
Error	fewer than 2 EI samples / all EI zero
Error	supports 1 and 2 at the same x / zero span



Tips and limits



Load shapes

Until fixed, Constant / Linear / Quadratic / Quintic all behave as Logistic.



Support 3

Takes a fixed share of the load; the deflection is zero only at supports 1 and 2.



Moment

The reaction moment cannot be negative.



Where it lives

Document	xSection (f8deedeb1fbd819a8fa20113), 'Stiffness Tools'
Tab	estimateDeflection (+ beam/estimateDeflectionSolver.fs)
Tests	none
Examples	Test2 studio: Estimate Deflection 1
Explainer	docs/explainers/xsection/xsection_explained.md, section 2.4
Related	EI and Cross Section, Estimate Stiffness, Estimate Deflection, Generate / Analyze baseline, Update profile